Arunabha Majumder

Mechatronics Engineer

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EDUCATION

PhD, Mechatronics and Robotics
Aalborg University | 2023 - present
Thesis expected to be finished in March
Study abroad: NUS Singapore, 2025

MTech, Mechatronics
CSIR-CMERI, India | 2020 - 2022

BTech, Mechanical Engineering
SIT (VTU), India | 2015 - 2019

Senior Diploma in Fine Arts (Painting)
Bangiya Sangeet Parishad | 2008, Distinction

SKILLS

SolidWorks (CSWA) | 3D printing
Fusion 360 (basic, last used 2019)
Motors, encoders, force/torque sensors
Arduino | LabVIEW | MATLAB/Simulink
PID control | Python (basic)
Qualisys motion capture and EMG
Hand sketching and painting
Illustrator | Photoshop | Procreate

AWARDS

Best Paper Award, IFToMM ISRM 2026
Best Major Project Award, 2019
TATA Technologies Prayatna grant

LANGUAGES

English (C1) | Bengali | Hindi

REFERENCES

Shaoping Bai, Professor
Aalborg University | sbh@mp.aau.dk

Lars Rosgaard Jensen, Associate Professor
Head of Section and Head of Labs, Aalborg University

Anderson de Souza Castelo Oliveira, Associate Professor
Materials and Production, Aalborg University

PROFILE

I build physical robot systems and get them working: mechanics, motors, sensors, wiring up the data side and the control loop. I like small teams where I can touch every part of the hardware.

EXPERIENCE

PhD Researcher, Robotics
Aalborg University, Denmark | *2023 - Present*

- Designed a hybrid shoulder exoskeleton (spring mechanism + 6 Nm motor) that uses over 25% less motor energy. Best Paper Award, ISRM 2026.
- Integrated the motor, encoder and force/torque sensors and measured stiffness on a physical setup.
- When a clutch kept breaking, swapped it for a simple bolt-locking clutch that was easy to make.
- Designed and modelled a novel 5-bar planar parallel robot with VSM; estimated its stiffness with less than 10% error and quantified human arm impedance.
- Ran data collection: EMG from 12 people and arm motion from 6 with Qualisys; implemented an Arduino trigger to sync robot motion with the recordings.
- Taught university courses and supervised Bachelor's students.

MTech Researcher, Soft Exosuit
CSIR-CMERI, India | *2020 - 2022*

- Fabricated and assembled pneumatic muscles; designed the outer PID loop of a neural network-based controller in MATLAB/Simulink and implemented it in LabVIEW to drive the muscle.

Bachelor's Projects
Siddaganga Institute of Technology, India | *2017 - 2019*

- Built an arm exoskeleton and exo glove for a paralysed arm: CAD, machining, assembly, testing.
- Built a Go-Kart and an Efficycle hybrid vehicle with student teams.